

Prompt Engineering

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Week 12: Ethics, Copyright, and Responsible Use

Why Responsibility Matters: Influence & Impact

- **Influence and Impact:** Prompt engineers design the inputs that shape AI behavior, directly influencing outputs that affect individuals, communities, and institutions.
- **Consequences of Neglect:** Ethical lapses can lead to significant harm, including reputational damage, erosion of public trust, legal liabilities, and reinforcement of societal biases.

Why Responsibility Matters: Balance & Sustainability

- **Balancing Act:** Responsible practice requires a delicate balance between pushing the boundaries of innovation and maintaining strict accountability for the system's outputs.
- **Long-term Sustainability:** Ethical AI development ensures the longevity and acceptance of AI technologies in society.

Weekly Learning Objectives

- **Risk Identification:** Learn to identify potential ethical risks and vulnerabilities in various prompting workflows and use cases.
- **IP Awareness:** Understand the complex landscape of intellectual property (IP) and copyright implications related to AI-generated content.
- **Framework Application:** Apply established responsible AI frameworks (e.g., NIST, EU AI Act) and organizational policies to prompt engineering tasks.
- **Ethical Design:** Design prompts that actively reinforce transparency, user consent, fairness, and explainability.
- **Crisis Management:** Develop robust escalation paths and incident response plans to handle AI failures or misuse effectively.

- **Bias, Fairness, and Inclusivity:** Ensuring AI systems do not discriminate and are accessible to diverse user groups.
- **Copyright, Licensing, Derivative Works:** Navigating the legalities of using copyrighted data and the status of AI-generated outputs.
- **Data Privacy, Consent, and Confidentiality:** Protecting user data and ensuring informed consent for data usage.
- **Responsible AI Principles:** Adhering to core principles such as transparency, accountability, safety, and reliability.
- **Incident Response and Red-teaming:** Proactively testing for vulnerabilities and having a plan for when things go wrong.

The Source of AI Behavior: Illusion & Data

- **The Illusion of Sentience:** A common misconception is that GenAI produces sentient thought; in reality, it creates content based on patterns learned from training data.
- **Data Provenance:** A significant portion of training data is sourced from the Internet, which serves as a "warehouse of misinformation," purposeful deception, and polarizing ideologies.

The Source of AI Behavior: The Bias Feedback Loop

- **The Bias Feedback Loop:**

- Uncurated web content introduces unintended biases (e.g., gender, racial, cultural) into AI systems.
- **Recursive Contamination:** As the web fills with AI-generated content, new models trained on this data may amplify existing biases, potentially leading to "model collapse".

Common Risk Areas: Bias & Misinformation

- **Bias and Discrimination:** Generating harmful or biased outputs that target protected groups or reinforce stereotypes.
- **Misinformation and Hallucination:** Creating fabricated citations, false facts, or spreading misinformation that can mislead users.

Common Risk Areas: Data, Automation & Security

- **Data Misuse:** Inadvertent exposure or misuse of personal identifiable information (PII) or sensitive corporate data.
- **Automation Bias:** The tendency for humans to over-rely on AI outputs without critical verification, leading to errors in decision-making.
- **Security Vulnerabilities:** Susceptibility to prompt injection attacks or jailbreaking attempts.

Step-Around Prompt Engineering (Adversarial)

- **Definition:** The practice of crafting inputs to deliberately bypass AI safety protocols and filters.
- **Purpose:**
 - **Research Tool:** Acts like "red teaming" in cybersecurity to expose underlying vulnerabilities and biases.
 - **Identifying Gaps:** It reveals how Internet-derived biases persist despite content filtering.
- **Risks:** If used maliciously, it can generate hate speech, disinformation, or aid in cyberattacks.

Risk Assessment Checklist

- **Stakeholder Impact:** Who might be harmed, directly or indirectly, by this output or workflow? Consider marginalized communities.
- **Safeguards:** What technical and procedural safeguards exist to detect and mitigate inaccuracies, bias, and toxicity?
- **User Awareness:** Are users clearly informed that they are interacting with an AI? Do they understand the system's limitations?
- **Monitoring and Auditing:** How are outputs monitored for quality and safety over time? Is there a feedback loop for corrections?
- **Data Provenance:** Do we have the right to use the data being processed? Is the data representative and fair?

Case Study: The "Woke" AI Scandal

- **The Incident:** In early 2024, Google Gemini attempted to be inclusive but generated historically inaccurate images (e.g., diverse Founding Fathers, Nazi soldiers).
- **The Cause:** An "overcorrection" in safety tuning aimed at mitigating bias.
- **The Lesson:** A tangible example of how "responsible AI" interventions can backfire if not tested properly.
- **Challenge:** Illustrates the difficulty of balancing bias mitigation with historical accuracy.

Understanding Copyright Basics: Core Question & US Stance

- **The Core Question:** Who owns AI-generated content?
- **United States Stance:**
 - **Human Authorship Requirement:** The US Copyright Office generally denies protection to works created solely by machines (e.g., *Thaler v. Perlmutter*).
 - **Hybrid Works:** Copyright may be granted if there is sufficient human creativity involved in selecting or arranging the AI output.

Understanding Copyright Basics: UK Stance & Litigation

- **United Kingdom Stance:** Unique in protecting "computer-generated works," assigning authorship to the person who made the arrangements for the work's creation.
- **Litigation Trends:** Major lawsuits (e.g., *NY Times v. OpenAI*, *Getty Images v. Stability AI*) focus on whether using copyrighted data for *training* AI constitutes fair use or infringement.

Prompting with Copyright in Mind

- **Avoid Direct Infringement:** Do not request the model to reproduce specific copyrighted text, lyrics, or images unless you hold the rights.
- **Encourage Originality:** Structure prompts to encourage original synthesis, analysis, and transformation rather than simple duplication.
- **Reference Material:** When grounding outputs in specific texts, ensure you have the appropriate license to use that material.
- **Citation and Attribution:** Explicitly instruct the model to cite sources when summarizing or transforming external content to maintain academic and professional integrity.

Case Study: The "Monkey Selfie" Precedent

- **The Story:** The US Copyright Office declared that works created by non-humans (like a monkey taking a selfie) cannot be copyrighted.
- **The AI Connection:** In *Thaler v. Perlmutter*, an AI system named DABUS was denied copyright for an image because it lacked a "human author".
- **The Twist:** "A Single Piece of American Cheese" was registered as a "composite work" because the human author argued their selection and arrangement constituted sufficient creativity.
- **Takeaway:** This highlights the subtle legal "grey areas" currently being fought over regarding human involvement.

Debate: The Ethics of "Style Mimicry"

- **Context:** Class action lawsuits by artists (e.g., Sarah Andersen, Kelly McKernan) against Stability AI and Midjourney for training on scraped images without consent.
- **Example:** Users generating "Ghiblification" images imitating Hayao Miyazaki's style, despite his known dislike of AI.
- **Discussion:** Where does the line lie between "inspiration" and "theft"? Is it ethical to prompt an AI to copy a living artist's style?

Reflexive & Responsible Prompt Engineering: Definition & Design

- **Definition:** A framework integrating ethical, legal, and social values into AI interactions.
- **The 5 Components of Responsible Prompting (1-3):**
 - ① **Prompt Design:** Systematically crafting instructions (e.g., using "chain-of-thought" reasoning) to maximize desired outcomes.
 - ② **System Selection:** Choosing the right model based on benchmarks and capabilities.
 - ③ **System Configuration:** Adjusting parameters like "temperature" to balance creativity and predictability.

Reflexive & Responsible Prompt Engineering: Evaluation & Management

- **The 5 Components of Responsible Prompting (4-5):**
 - ④ **Performance Evaluation:** Using metrics and human assessment to check output quality and consistency.
 - ⑤ **Prompt Management:** Documenting and version-controlling prompts for accountability.

Embedding Guardrails

- **Safety Disclaimers:** Include clear disclaimers about the AI's nature and scope limits within the system prompt.
- **Refusal Mechanisms:** Instruct models to politely refuse sensitive, harmful, or out-of-scope requests (e.g., "I cannot assist with that request").
- **Uncertainty Expression:** Encourage the model to express uncertainty when confidence is low: "If the answer is not in the context, state that you do not know."
- **Policy Blocks:** Use specific instructions for sensitive topics: "Do not provide medical diagnoses; instead, recommend consulting a qualified healthcare professional."

Transparency and Accountability

- **Disclosure:** Clearly disclose AI assistance in all outputs where required by law or ethical standards.
- **Logging and Auditing:** Maintain comprehensive logs of prompts, context, and generated outputs for audit trails and debugging.
- **Documentation:** Document the rationale behind prompt design decisions and any human oversight actions taken.
- **User Guidance:** Provide users with clear guidance on system limitations, review steps, and contact information for reporting issues.

Discussion: The "Illusion of Transparency"

- **Topic:** When we ask AI to "think step-by-step" (Chain-of-Thought), is it telling the truth?
- **Research:** Suggests explanations might be an "illusion of transparency"—plausible justifications that don't match the model's actual mathematical path.
- **Implication:** Can we trust AI explanations in high-stakes fields (law, medicine) if the logic path is fabricated to sound convincing?

Tangible Tip: "Step-Around" Prevention

- **Advice:** Responsible prompting isn't just about good text; it's about checking for "Refusal" and "Hallucinations."
- **Technique:** Use "Chain of Thought" prompting to force the model to evaluate its own biases.
- **Example Prompt:** "Plan verification questions to fact-check this draft" or explicitly ask the model to review its output for specific biases before finalizing.

- **NIST AI Risk Management Framework (AI RMF):**
 - **Govern:** Cultivate a culture of risk management.
 - **Map:** Establish context and identify risks related to that context.
 - **Measure:** Assess, analyze, and track identified risks.
 - **Manage:** Prioritize and act upon risks based on projected impact.

- **The EU AI Act:** The world's first comprehensive AI law, adopting a risk-based approach.
 - **Unacceptable Risk (Banned):** Social scoring, manipulative cognitive techniques.
 - **High Risk (Regulated):** Critical infrastructure, education, employment tools.
 - **Transparency Risk:** Chatbots and deepfakes must clearly disclose that content is AI-generated.

Incident Response

- **Monitoring:** Actively monitor for policy violations, biased outputs, or user complaints using automated tools and human review.
- **Escalation:** Establish clear channels to escalate incidents to responsible owners (legal, ethics, engineering) quickly.
- **Post-Incident Review:** Conduct thorough post-incident reviews to identify root causes, implement remediation, and communicate findings.
- **Iterative Improvement:** Update prompts, training data, and governance artifacts based on lessons learned from incidents.

Regulation Example: The "Social Scoring" Ban

- **Context:** The EU AI Act bans "social scoring" (classifying people based on behavior/status) and real-time biometric identification in public spaces.
- **High Risk:** Educational and vocational training AI is classified as "High Risk".
- **Debate:** Should schools be allowed to use AI for "emotional surveillance" or predicting failure? Do you agree with these bans?

AI in Education: Plagiarism & Pitfalls

- **Plagiarism Definition:** Submitting AI-generated text as one's own work is generally considered plagiarism because it lacks the student's own analysis and critical thinking.
- **Common Pitfalls:**
 - **AI Hallucinations:** AI often fabricates academic citations that do not exist.
 - **Paraphrasing Tools:** Using AI to rewrite sources without citation is still academic misconduct.

AI in Education: Detection & Best Practices

- **Detection:** Tools like Turnitin analyze sentence patterns and word probability, though they can produce false positives.
- **Best Practices:** Use AI for brainstorming or grammar checks, but manually verify all citations and ensure the core arguments are your own.

Future Outlook & Recommendations

- **Organizational Responsibility:** Companies need "Responsibility by Design"—embedding ethics into the development process rather than treating it as an afterthought.
- **Transparency:** Developers should maintain detailed documentation of prompts and training data sources to ensure accountability.
- **Human-in-the-Loop:** Hybrid collaboration, where humans oversee AI outputs, is essential for copyright protection and risk management.

Key Takeaways

- **Integral Practice:** Ethics and compliance are not optional add-ons but are integral to the practice of professional prompt engineering.
- **Protective Design:** Design prompts that actively protect users, respect intellectual property rights, and promote system transparency.
- **Accountability:** Document all policies, incidents, and improvements to create a culture of accountability.
- **Continuous Commitment:** Responsible AI is an ongoing commitment that requires constant vigilance and adaptation to new challenges.

Thank You!

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Questions?